ORIGINAL PAPER

Boltz-2 Modeling Suggests a Structural Basis for Inhibition of the Multidrug Pump ABCB1 by Repurposed and Plant-Derived Drugs

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I. Abstract

The ATP-binding cassette subfamily B member 1 (ABCB1), also known as P-glycoprotein (P-gp), is a highly clinically relevant ATP-dependent efflux pump that exports chemotherapeutics and other toxic xenobiotics out of cancer cells. Consequently, competitive ABCB1 inhibition by small-molecule drugs can aid in overcoming multidrug resistance (MDR) and can heighten the intracellular concentration of chemotherapeutics, increasing overall therapeutic efficacy. Although many synthetic and naturally-derived ABCB1 inhibitors have been identified and tested clinically, their lack of specificity and severe toxicity makes them unviable for disease treatment. Consequently, to broaden the landscape of potential inhibitors, we employed Boltz-2 (v1.0.0) and PLIP to screen nine ABCB1 small-molecule systems where unclear protein-ligand binding mechanisms exist to validate their hit discovery against known, clinically useful ABCB1 inhibitors. Our *in silico* screenings identify Lamellarin I, Lobeline, and Alvocidib as novel hits for ABCB1 inhibition, while Amiodarone, Dofequidar, and Lamellarin I were the top three candidates that combine sub-micromolar predicted IC_{50} with both high confidence metrics (>0.778) and high hit-discovery (>60%). Taken together, these findings position Amiodarone, Dofequidar, and Lamellarin I as promising ABCB1 inhibitor candidates warranting structural experimental and clinical validation, offering a potential new avenue for overcoming ABCB1-mediated MDR in cancer therapy.

II. Introduction

The ABCB1 gene encodes P-gp/the ABCB1 transporter, a membrane-associated ATP-dependent transporter that exports xenobiotics out of human cells. ABCB1 is commonly unregulated across multiple cancer types; therefore, as a consequence, ABCB1 overexpression has been linked to the development of chemotherapy resistance. Current bottlenecks in ABCB1 inhibitor development and clinical validations include severe systemic toxicity, transporter polyspecificity, and unpredictable pharmacokinetic alterations. Inhibiting ABCB1 in areas such as the liver, intestine, kidney, and blood-brain barrier can change normal protection, absorption, distribution, and elimination of drugs and xenobiotics. This itself can expose healthy tissues, including the CNS, to higher drug concentrations.

Moreover, ABCB1's flexible and hydrophobic binding cavity has the ability to recognize a wide range of substrates, complicating the design of an effective inhibitor that is competitive, potent, selective, and non-toxic. Consequently, hypothesis screening of a vast array of small molecule inhibitors through computational methods enables an accelerated search for specific ligands that may prove beneficial in a clinical setting. By leveraging these computational methods, we're able to structurally predict ABCB1-ligand complexes at atomic resolution – ultimately streamlining the pipeline between target identification and clinical approval.

III. Results

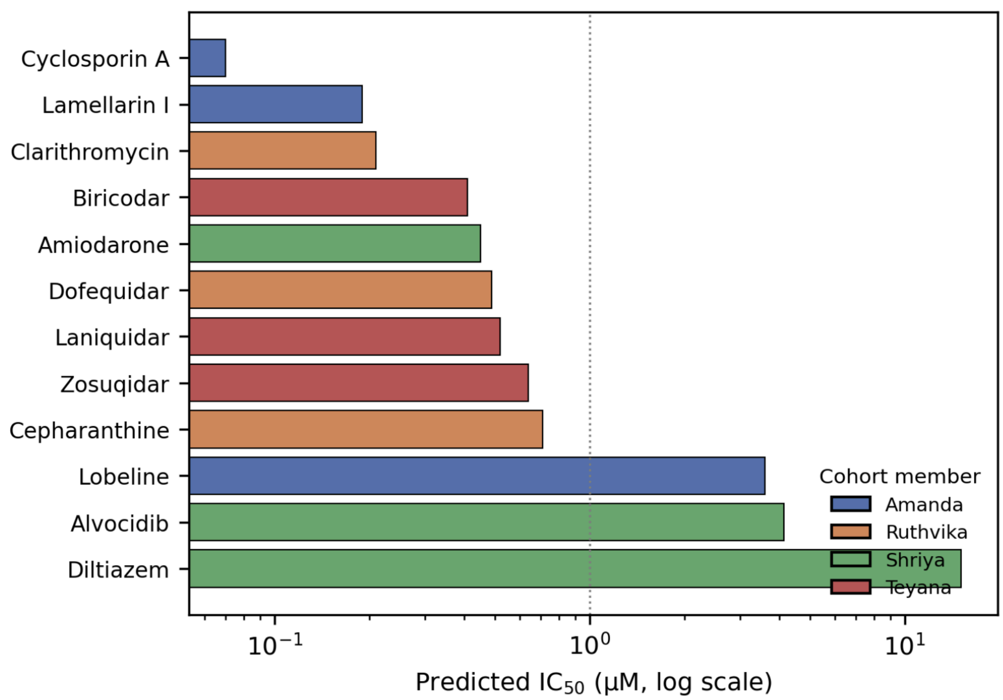


Figure 1. Predicted IC_{50} Values Per Drug Candidate. Predicted IC_{50} (half-maximal inhibitory concentration) values generated by Boltz-2 (v.1.0.0) [Boltz2-Notebook]. Screenings conducted by each team member are color-coded, respectively. Blue = Amanda; Orange = Ruthvika; Shriya = Green; Teyana = Red. Figure generated using Matplotlib.

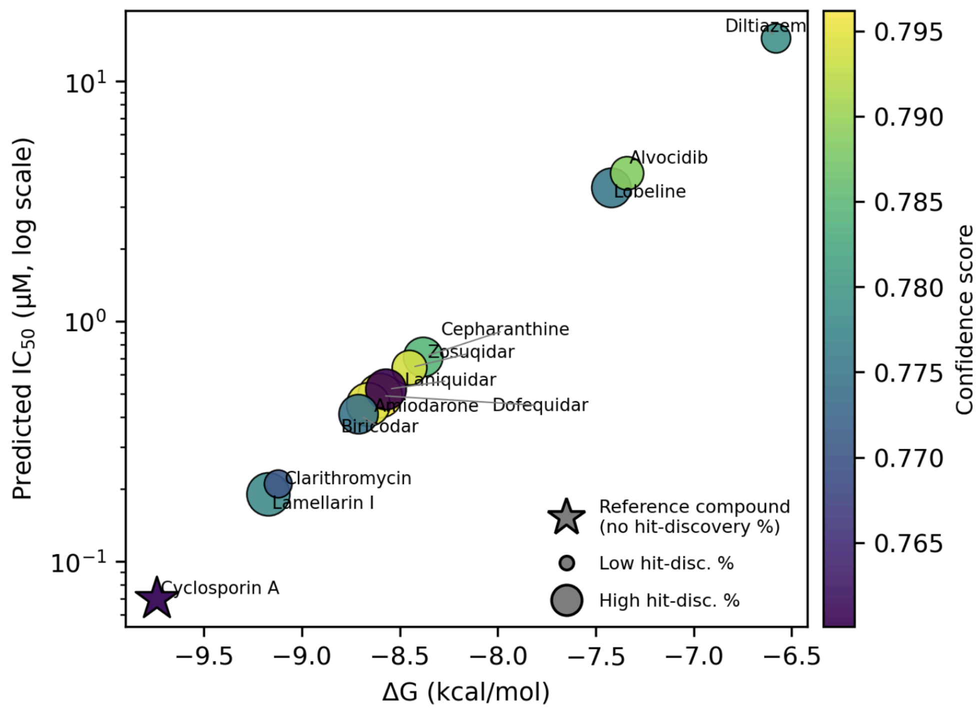


Figure 2. Predicted IC_{50} vs Binding Energy. Predicted IC_{50} and binding energy (kcal/mol) values were generated by Boltz-2 (v.1.0.0) [Boltz2-Notebook]. Star indicates the reference compound; bubble size indicates hit-discovery % as calculated by Boltz-2. Figure generated using Matplotlib.

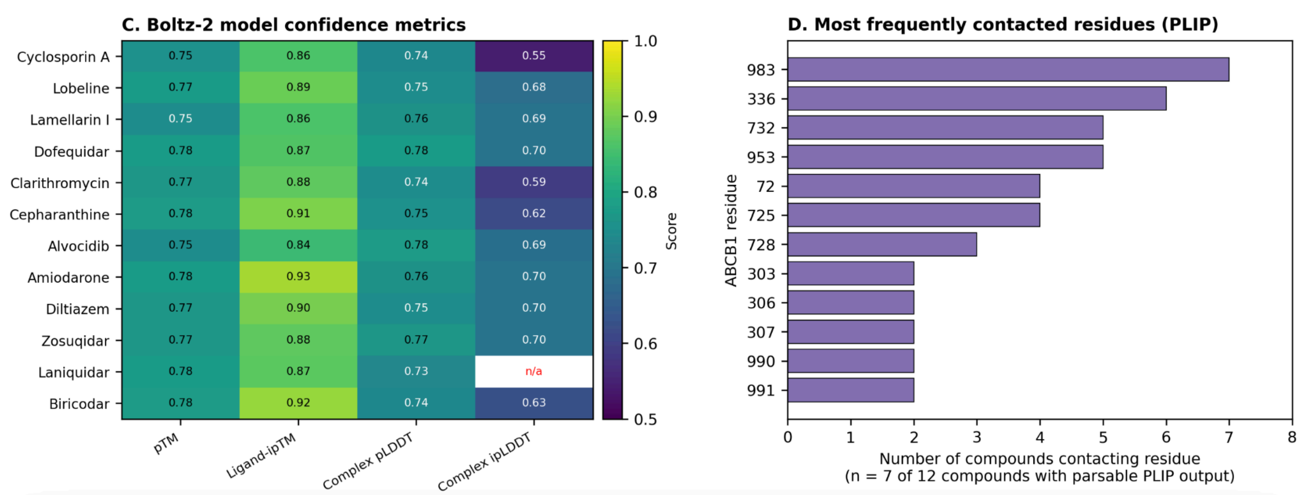
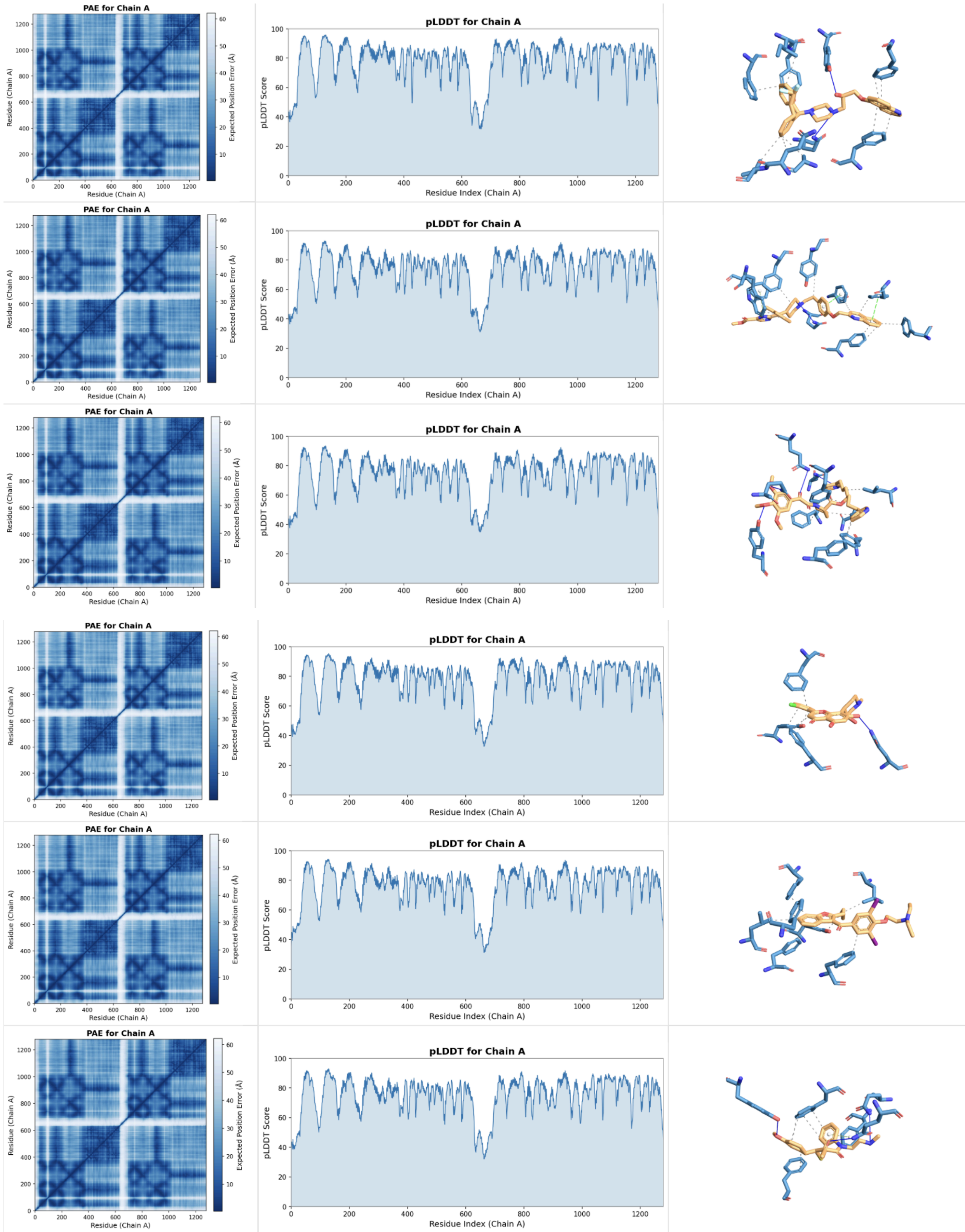


Figure 3. Boltz-2 Model Confidence Metrics and Most Frequently Contacted Residues. Heatmap of pTM, Ligand-ipTM, Complex pLDDT, and Complex ipLDDT per small molecule candidate (Left). Bar chart of the most frequent residues where non-covalent interactions exist between ABCB1 and screened compounds as per PLIP analysis (Right). Figure generated using Matplotlib.


PAE for Chain A
Residue (Chain A)
Residue (Chain A)
Expected Position Error (Å)
pLDDT for Chain A
pLDDT Score
Residue Index (Chain A)
PAE for Chain A
Residue (Chain A)
Residue (Chain A)
Expected Position Error (Å)
pLDDT for Chain A
pLDDT Score
Residue Index (Chain A)
PAE for Chain A
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Residue Index (Chain A)
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Expected Position Error (Å)
pLDDT for Chain A
pLDDT Score
Residue Index (Chain A)

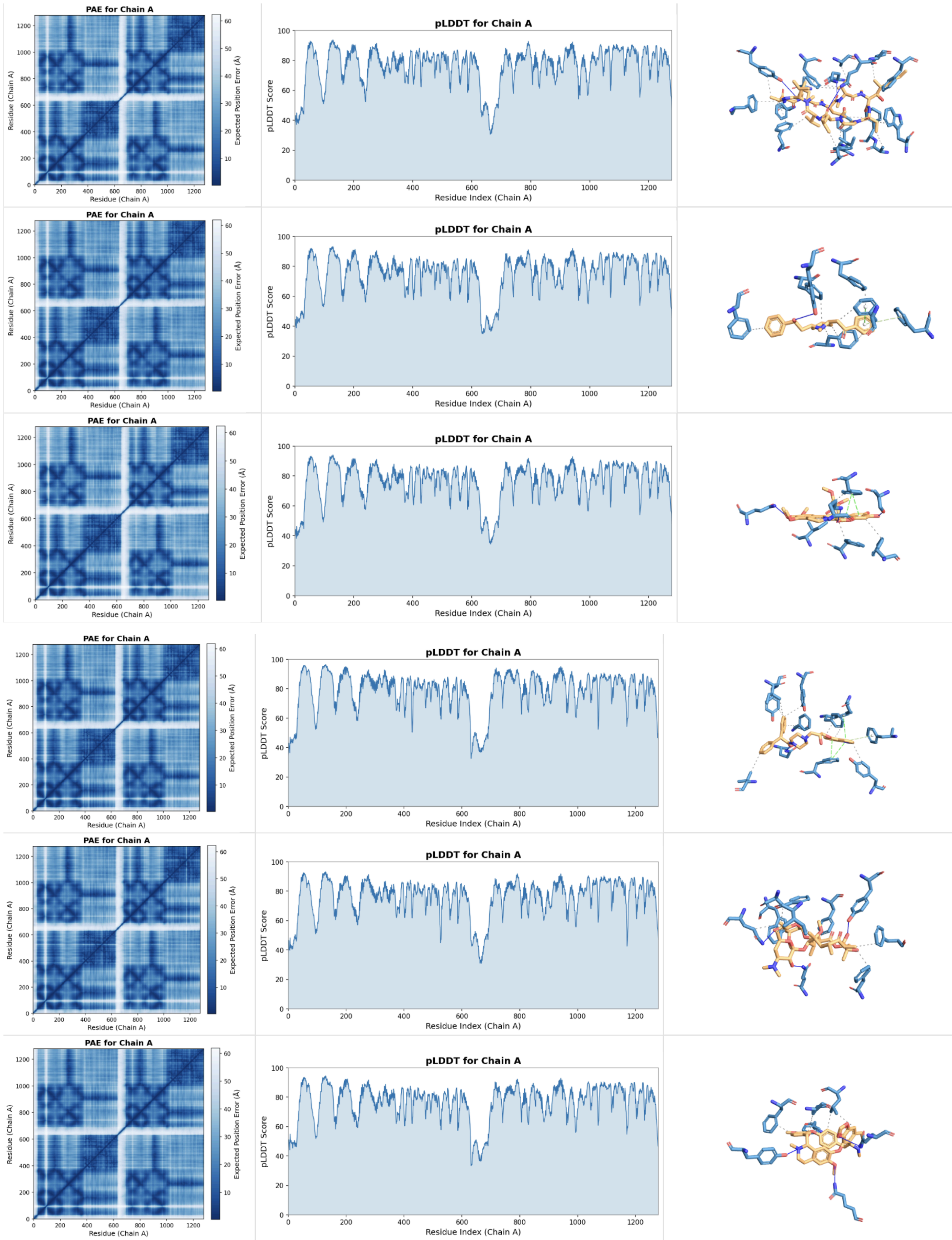

Figure 4. ABCB1-Ligand Complex Predictions by Boltz-2 and PLIP. Predicted Aligned Error (PAE) (Left), predicted Local Distance Difference Test (pLDDT) (Center), and PLIP non-covalent interaction

analysis (Right) for all twelve ABCB1 (Chain A)-Ligand structure predictions. In descending order, the candidate compounds for $PAE_{CHAIN\ A}$, $pLDDT_{CHAIN\ A}$, and PLIP analysis per row are: Zosuqidar, Laniquidar, Biricodar, Alvocidib, Amiodarone, Diltiazem, Cyclosporin A, Lobeline, Lamellarin I, Dofequidar, Clarithromycin, Cepharanthine.

IV. Discussion

Predicting protein-ligand interactions using Boltz-2 can be a powerful tool for hypothesis generation and the prioritization of candidate compounds; however, these predictions ought to be interpreted as computational prioritization solely. Biomolecular prediction software cannot guarantee clinical potential without thorough experimental validation, as predictions and confidence metrics heavily depend on pre-existing structural information on target proteins.

However, with our preliminary structure-based virtual screening of twelve candidate ABCB1 inhibitors, Boltz-2 yielded predicted IC_{50} values spanning 0.07-15.12 µM and binding energies of -6.58 to -9.74 kcal/mol, with confidence scores ranging from 0.76-0.80. Amiodarone (IC_{50} 0.45 µM, ΔG -8.66 kcal/mol, confidence 0.794), Dofequidar (0.49 µM, -8.60 kcal/mol, 0.796), and Lamellarin I (0.19 µM, -9.17 kcal/mol, 0.778) ranked highest by combined potency and model confidence, performing comparably to the reference third-generation inhibitors zosuquidar, laniquidar, and biricodar (IC_{50} 0.41-0.64 µM), with Lamellarin I surpassing all three.

Lower-ranked candidates diltiazem (15.12 µM) and alvocidib (4.14 µM) delineated the weak end of the screening choices despite comparable confidence scores (0.779 and 0.789). Residue 983 was contacted by all seven compounds via PLIP analysis, including Amiodarone, Dofequidar, and Lamellarin I, suggesting convergence on a shared residue within the ABCB1 substrate-binding pocket. Unlike amiodarone and dofequidar, both accepted ABCB1 modulators, Lamellarin I's predicted ABCB1 interaction has no prior literature precedent, suggesting Lamellarin I may be a novel ABCB1 inhibitor if experimental studies demonstrate similar results.

V. Methods

A library of both naturally-derived and repurposed compounds was constructed based on prior original articles and relevant review articles using combinations of the terms ABCB1 inhibitors, P-glycoprotein inhibitors, MDR1 inhibition, ABCB1 efflux, multidrug resistance reversal, and ABCB1 ATPase. To predict our protein-ligand complexes, Boltz-2 was run on Google Colab using the Boltz-2 Notebook. The following parameters were used to run Boltz-2 on the NVIDIA A100 Tensor Core GPU.

Use Potentials	Enabled
Recycling Steps	7
Sampling Steps	200
Diffusion Samples	10
Step Scale	1.638
Max MSA Sequences	8192
Subsample MSA	Disabled
MSA Pairing Strategy	greedy

Following prediction generation by Boltz-2, output PDBs were uploaded into Protein-Ligand Interaction Profiler (PLIP) using the protein-ligand interactions (default) setting to analyze non-covalent ligand-residue interactions.

VI. References

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